

SUMMARY

I'm a full stack software engineer specializing in the Web3 stack, mostly TypeScript and Solidity. With over 11 years of professional experience, I've had the opportunity to work on bleeding edge products in Web3, including having been part of the Zero Knowledge University sponsored by Harmony where I learned how to write ZKPs, have spent some time digging into smart contract auditing, and have taken a winning ETHDenver hackathon project into a startup. In Web2, I even got to work with Vitalik Buterin directly by implementing dark mode on his blog. I've worked on projects that run the gambit from federal government tools, private business enterprise applications, custom websites for digital marketing agencies, and freelance consultant. In my next role, I am looking forward to continuing to move the needle in Web3 by being innovative and contributing to building the future of the decentralized web. I hope to work alongside a team that is as passionate about changing our internet landscape as I am!

EXPERIENCE

NASD, Inc (Noble) Frontend Web3 Engineer

May 2025 - Present • United States (Remote)

- Noble is a purpose-built blockchain designed exclusively for stablecoin issuance across the Cosmos/IBC ecosystem. Noble builds liquidity infrastructure for asset issuers to participate in the Interchain economy, and it is responsible for issuing ~389M (as of Aug '25) in USDC across IBC chains.
- I help maintain the frontend dapps for the Noble product offerings - Express, Dollar, and Points.
- I've helped convert disparate frontend repos into a Turborepo monorepo and have significantly improved the DX since joining the team. I've implemented a robust set of Github Actions to build, test, lint, and deploy the different applications. I DRY'd up all of the tsconfig, eslint, prettier, and TailwindCSS configs for reuse and consistency.
- I've started building a component library for use across the three different frontends to simplify development and make things more consistent with a focus on best practices and a11y as well as intuitive API/props/variants.
- Tech Used:** Next.js, React, Turborepo, Typescript, TailwindCSS, Cloudflare Workers/Pages, Github, Cypress

Hypotenuse Labs Web3 Engineer / Consultant

Dec 2024 - Present • United States (Remote)

- Hypotenuse Labs is a boutique development firm specializing in web and mobile applications in the spaces of blockchain/Web3, AI, and traditional Web2 products. The firm focuses on staffing senior-level engineers and building new and innovative products for clients, who are generally in the initial product ideas or a funded startup. The firm positions itself within the market by providing high quality code and best engineering practices.
- I am one of about 30 engineers on the team, and I specialize in blockchain/Web3 with a focus on frontend development, TypeScript backend services and APIs, blockchain event indexers, database development, and smart contracts in Solidity.
- I work in a client-facing position where I am involved in many different points of a project's life. A greenfield project might look like this: Client comes with an idea and need a project scope, I deconstruct the idea into components, outline what each will do and what it will be built with, outline the order of operations and deliverables with certain milestones, outline how many engineers would be needed for the project, and estimate the number of hours it will take. If they take the bid, then we are staffed on the project and have weekly meetings with the client while executing on the proposal.
- I conduct full stack interviews with candidates. I present a problem set in Typescript/React/SQL and provide a report on the candidate's performance on both technical and personal communication levels.
- I've had the opportunity to be exposed to a lot of new ideas through scoping projects and have helped contribute to several projects that are under contract, mostly lending my expertise in the in the frontend capacity and backend Typescript services.
- Project - Sushiswap**
 - As the sole frontend engineer on a team of 4, I worked in the Sushiswap UI codebase to help implement support for a new blockchain.
 - I added wallet support for the new network, added the network configuration objects, Typescript bindings for the native structs, and created the necessary hooks for submitting both generic and specific transactions to the network and to deployed smart contracts.
 - I was able to fork the existing Sushiswap interface repo and work within the existing application structure and design patterns, ensuring that my code would be easy to maintain after handoff and there would be very little context switching from the rest of the codebase.
 - Implemented pages and functionality for token swaps, exploring pools, viewing pool details, and providing liquidity to pools, all talking to the newly deployed smart contracts the rest of the team built.
- Tech Used:** TypeScript, Nest.js, Next.js, React, Turborepo, Subsquid, Ponder.sh, TypeScript, Solidity, Foundry, Postgres, MongoDB, Figma

Alkimiya Lead Full Stack Engineer

Jan 2024 - Dec 2024 • United States (Remote)

- Alkimiya is a defi protocol built on Ethereum that allows for hedging against the rise and fall of the average B TC transaction fees over a range of dates. As part of a small team of only a few engineers, I contribute to the full stack, and am responsible for a large amount of backend code.
- I led high-level design decisions and the overall direction of many parts of the stack.
- I designed database schema for the offchain data and backend infrastructure for the V1 launch of the protocol.
- I helped design the event schema for the oracle and pools smart contracts to pair better with offchain ingestion.
- I implemented a full feature GraphQL and REST server in Nest.js, which includes lots of custom validation, authorization for protected routes, ample logging, and monitoring with Sentry.
- I've built several blockchain indexers using Subsquid and Ponder.sh to capture onchain events and ingest the log data into our offchain data stores. I also designed the database schema for these as well.
- I've contributed to and help scale up the Next.js React frontend by preparing it to be able to manage multiple networks and markets in a DRY and maintainable way
- Help set up AWS infrastructure including EC2, RDS, S3, Route 53, and ELB.
- Tech Used:** TypeScript, Nest.js, Next.js, React, Subsquid, Ponder.sh, Go, Solidity, Foundry, Postgres, MongoDB

SKILLS

Programming Languages

JavaScript, TypeScript, Solidity, C#, Go
SQL, GraphQL
HTML, CSS, YAML, XML

Frameworks & Tooling

Foundry, Hardhat, Viem, Wagmi, Ethers.js, Web3.js
Slyther, Echidna, Mythril, Certora Prover
Chainlink, IPFS, Infura, Alchemy, The Graph, Subsquid
.NET, Node, Next.js, Express, MongoDB, Mongoose, Apollo,
React, Vue, Next.js, Nuxt.js, Redux, Zustand, React Native
Jest, Mocha, React Testing Library, Enzyme, Cypress
Docker, Kubernetes, AWS

Certifications

Certified ScrumMaster
ScrumAlliance Certificate ID #355624

Education

SeedPaths
Aug 2014 - Oct 2014 • Denver, CO
8-week intensive software development bootcamp
learning full stack, object-oriented programming in
C#, .NET, JavaScript, jQuery, HTML, CSS, SQL, Git

Louisiana State University
Aug 2008 - May 2011 • Baton Rouge, LA
Landscape Architecture & Communication Studies
Learned AutoCAD, Photoshop, InDesign, & Illustrator

Entrepreneurial

Althea Web Services, LLC
Jun 2016 - Present • Denver, CO
I'm the sole proprietor of this business, primarily set
up for freelancing, consulting, and liability protection

Polyecho Labs, LLC
Feb 2022 - Dec 2022 • Denver, CO
Incorporated the LLC in regards to Arbor Labs with
two other co-founders in a general partnership in
efforts to raise funding and deliver digital products

Drew Cook

Full Stack Software Engineer

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Aave Companies Senior Full Stack Engineer

Jul 2022 - Feb 2023 · London, UK (Remote)

- Providing full stack engineering support on the interface team for the Aave Protocol. We owned the client application at app.aave.com, which included a Next.js user interface, client-side Web3 helper libraries, a backend Node.js API, a MongoDB database, and middleware libraries that would interact and make calls to the Solidity smart contracts for the protocol. I made code contributions to all of these layers during my time at this role.
- I helped improve the open source repository documents which outlined how developers could clone or fork our codebase and build their own user interface. I authored the README.md and CONTRIBUTING.md documents that outlined how developers could contribute effectively to the open source repository, which were lacking descriptive information when I joined.
- I helped design an open-source tracking system that we used for managing open issues and pull requests from the community as well as how we could use Github internally for tracking internal feature work, bugs, and improvements to the codebase.
- I advocated for and set up automated pre-commit and pre-push hooks which ran a complex suite of linting and code style configs to keep all files consistent prior to them making their way into the remote branches. This also included using Commitlint, which enforced consistent commit messages to keep git history clean and descriptive.
- I advocated for and set up a unit testing harness as well as CI/CD automation for running these which did not exist beforehand, and advocated to the team to start writing unit tests for any utilities or non-component-based code that we would write, which did not exist prior to my joining.
- I participated in cross-team discussions about what the interface means for an open source Web3 protocol and how it fits into the broader scope, and helped to author community messaging about that to create clarity.
- I worked alongside other interface engineers, designers, product owners, smart contract engineers, dev ops, and quality assurance engineers, and the C-level executives on a daily or as needed basis to ensure we were aligned in our code execution and meeting our overall objectives within the Aave Companies.
- I triaged and prioritized open source issues and pull requests from the community and engaged directly with open source devs by performing peer reviews and having discussion.
- I collaborated with BGD to align objectives and code architectural changes or concerns that would improve the interface.
- I collaborated with other web3 teams on code integration while also guiding our code style preferences for their engineers to implement the different features using their products and technologies within our codebase.
- **Tech Used: TypeScript, Next.js, Node, React, IPFS, MongoDB, Jest, Cypress, Solidity, Tenderly, Figma, Github**

Arbor Labs Co-Founder / CTO

Feb 2022 - Dec 2022 · United States (Remote)

- Arbor Protocol is a Web3 protocol and product centered around generative music NFTs with royalties baked in. Users can upload stems (individual instruments within a track) to an Arbor musical "project", where others may listen and playback. Other collaborators can add stems onto the project to build up a song, which can then be minted as an NFT. All stems and NFT metadata is stored on IPFS via NFT.storage at Protocol Labs.
- The mint price is evenly distributed among all collaborators, and the NFT can be listed for sale on the secondary market. When a secondary sale is made, the seller receives 90% of the proceeds and the other 10% is split evenly among the collaborators in the form of royalties, all on-chain.
- This project started at the ETHDenver IRL hackathon which one first prize from Filecoin. We went on to build this during the ETHDenver virtual hackathon the following month and won top placements in the NFT category, top 30 overall projects, and top 5 projects from celebrity judges. We're using the winnings to fund our treasury and support onboarding new producers and artists onto the platform.
- Delegation of feature work, peer reviews, code style guidance, and mentorship through engagement with open source engineers being paid through Coordinape and a the Harmony zkDAO, which consists of engineers that graduated with me during Zero Knowledge University, a Harmony-backed program.
- I have been responsible for building both the frontend and backend for the web application, the Solidity smart contracts, as well as oversight onto technical decisions for the project.
- **Tech Used: IPFS, Solidity, Next.js, React, MongoDB, Jest, Python, Semaphore, ZK Proofs, Github**

Charter Communications Senior Design Technologist

Aug 2021 - Jun 2022 · Greenwood Village, CO

- Providing technical insight and engineering support on the core design systems team. We author and make design decisions that impact the entire suite of Spectrum products at Charter, and I build the tech around it. I owned the development and direction of a new technical documentation site outlining the design system principals and the implementations of them across the frontend JavaScript framework implementations of it, written in TypeScript and Vue 3.
- I also optimized the web performance and accessibility of our design system component libraries and help shape the developer experience for all product engineers utilizing our components across React, Angular, Vue, and vanilla JavaScript component libraries.
- This role was a good mix of technical development and interfacing with the developers that were consuming the libraries and design tokens that we created, how to use them, which ones to use based off of specificity, and how to implement the libraries into their existing Angular, React, Vue, or vanilla JavaScript applications, including TV.
- **Tech Used: TypeScript, Vue3, React, Angular, VuePress, Sass, Gitlab, Docker, Kubernetes, AWS, Jira**

Architect of the Capitol Senior Full Stack Engineer

Sep 2020 - May 2021 · Washington DC (Remote)

- I came into an existing monolithic app migration that is used once every US election cycle (two years) from .NET 4.x to .NET Core 3.x and had adapted into a new team that was unfamiliar with the app's use case. Since I worked on the app previously, I possessed all the historical knowledge and context and provided insight to the team on data relationships, why decisions were made around code architecture, and to clarify expected application use cases. (Please see older experience below)
- I helped speed up future schema migrations by writing many different SQL scripts to be used for every election cycle in the future for easier migrations, since congressmen and congresswomen could rotate in or out of office.
- I had a direct impact on the onboarding process for new hires and developers across the broader organization new to the codebase, by authoring all documentation for the new front end architecture and defined best practices and principles to follow.
- I impacted the efficiency of stakeholder feature demos and QA testing by creating SQL scripts to mimic specific application states, which were used by both developers and QA for testing bugs and setting up feature demos, resulting in a more seamless workflow and a happier client during demos.
- I optimized and improved the code quality by owning the rewriting of the client application from an MVC view-based approach to a SPA and upgraded to React 16.5, rewriting all class-based components to functional components with hooks, implemented the Context API for global storage and authored over 330 individual tests across 110 test suites to reach 80% code coverage. This gave me the opportunity to catch all my oversights and apply newfound knowledge in React I gained over the years.
- **Tech Used: .NET, C#, SQL, React, Jest, SignalR, Hangfire, React Testing Library, Sass, Bootstrap, ESLint, Prettier, Team Foundation Server**

Gloo Senior Frontend Engineer

Jun 2019 - Sep 2020 · Boulder, CO

- I worked on the Accounts/Registration team for onboarding new users and organizations registering with the Gloo platform. We solved the problem for tracking where a new user's external origin was, and through logic, determining their proper destination to be, scaling for an SSO model across the platform's host of products. Our UIs had a direct impact on every user, optimizing the conversion rates for successful registrations through efficient user flows, registrations, password resets, verification codes via email, etc.

Gloo (continued)

- As the only full-time web engineer, I was responsible for ensuring our Gloo Web team policies and practices were being upheld through code reviews, guidance, and leadership.
- I contributed to the internal Gloo Web team by authoring two NPM packages to implement custom Google Analytics and Facebook tags within our Next.js applications.
- I sped up our onboarding process for web contributors by spearheading a Naming Conventions Best Practices doc on front end code for React, Apollo GraphQL, Rosie, Jest, Next.js, etc. authoring it, coordinating with team members for input, and getting everyone to sign off on it, which we later adopted and included our onboarding docs for new hires or contractors.
- Tech Used: Typescript, Next.js, Apollo GraphQL, Node.js, Express, CircleCI, Heroku, Docker, Kubernetes, Jest, Cypress, React Testing Library, Material UI, Formik, Rollbar

Architect of the Capitol Full Stack Engineer

Mar 2018 - Jun 2019 · Washington DC (Remote)

- Project - Compass
 - I led the front end development during a redesign of the internal SharePoint intranet for employees about organizational news, events, and other important information.
 - I optimized and reduced the time and stress for content creators within the organization by planning, developing, and launching an internal tool for creating custom SharePoint content, catered to the site. This was used by internal employees that were responsible for creating events, promotional articles, image uploads, etc within Compass.
 - I helped mentor other team members on pieces of code / development concepts when the time was needed.
- Project - Senate Suites
 - I owned the client application development during an enterprise app redesign in .NET, migrating from an MVC architecture with views written in Razor to views written in React.
 - I leveraged Redux for state management and Webpack for modular bundle sizes, ensuring only proper bundles were loaded for relative pages within the app.
 - I ushered in and implemented a testing strategy in Jest and Enzyme, resulting in a test coverage of 78% of all client code. Previously, there were no tests implemented at all.
 - I supported migrating features from ColdFusion into .NET, and implemented WebAPIs for a RESTful approach, resulting in a faster and less buggy app.
 - I worked directly with stakeholders of the project, organizing and planning feature work, tech debt, and long term trajectory, while building up client communication skills.
 - Tech Used: C#, .NET, SQL, React, TypeScript, Redux, Jest, Enzyme, jQuery, SharePoint, CAML, Webpack, ESLint, Less

Evolver, Inc. Full Stack Engineer

Apr 2017 - Mar 2018 · Centennial, CO

- I dramatically improved the user experience for over 10,000 users by optimizing client code and refactoring a .NET MVC web app with jQuery to using React and Redux. I brought in shipping confidence for the app and improved our deployment pipeline by implementing a testing harness in Jest and Enzyme covering over 75% of client code, configuring linting with ESLint and formatting with Prettier for overall code quality.
- Tech Used: C#, .NET, React, ES6 JavaScript, CSS, Sass, Webpack, Redux, Jest, Enzyme, Prettier, Jira, jQuery

Crawford & O'Brien Lead Web Developer

Sep 2015 - Apr 2017 · Tempe, AZ (Remote)

- I served as the lead developer for a digital marketing agency performing any and all tech related tasks for over 40 clients, including but not limited to: development of WordPress and Shopify sites with custom themes, continued client support requests, DNS and content migrations, emails, security, backups, site speed optimizations, SEO, etc.
- I built up personal communication skills by working directly with the clients to find out their needs and accomplishing the tasks needed to meet those. I was the main point of contact for direct requests for their websites, and I interacted with them on a 1:1 basis.
- As our department lead, I directed and managed 3 junior developers by delegating tasks and implementing, documenting and overseeing our processes of web development.
- Tech Used: WordPress, PHP, Shopify, HTML, CSS, JavaScript, Google Analytics, WooCommerce, Yoast SEO, Trello

Access Now Web Developer / Designer

Dec 2014 - Aug 2015 · Greenwood Village, CO

- As a contributor, I designed and built dynamic, responsive websites and custom logos and marketing materials for a digital marketing agency. I worked directly with clients.
- Tech Used: WordPress, PHP, HTML, CSS, JavaScript, MySQL, cPanel, Plesk, Adobe Photoshop, InDesign, and Illustrator